

Stack Overflow Survey Analysis

IBM Data Analyst Capstone Project

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix



EXECUTIVE SUMMARY



- Objective behind interpreting and analyzing data.
- Methodology description.
 - Data collecting.
 - Data analysis.
 - Data Visualizations.
- Display of outcomes using visual aids such as graphs and trend analysis.
- Interpretation of key findings and their significance based on the presented results.
- Summary conclusions drawn from the conducted research.



INTRODUCTION



- Stack Overflow conducts the world's most extensive and detailed yearly survey of programmers globally.
- The findings may not equally reflect all segments of the developer community.
- Patterns used to forecast future directions in the developer landscape.
- Profiling developers from different parts of the world.



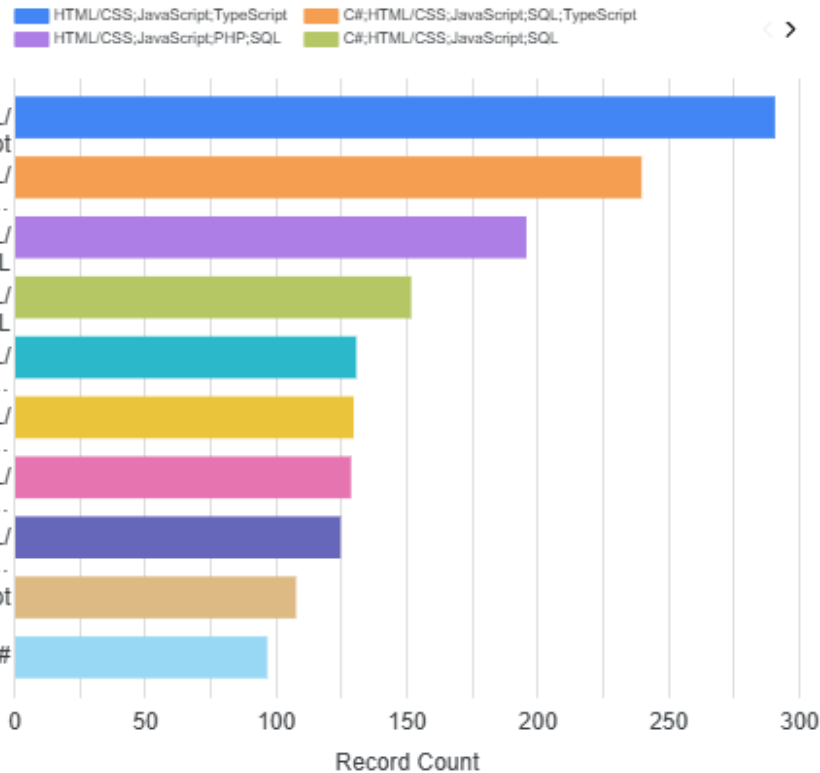
METHODOLOGY



- Collect Data
 - Web Scrapping
- Data Wrangling
- Data Analysis
- Data Visualization
 - Illustrate how the data is spread, connected, structured, and compared
- Dashboard



Current Year



Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- C#; HTML/CSS; JavaScript; SQL; TypeScript became the leading languages
- C# has significant growing

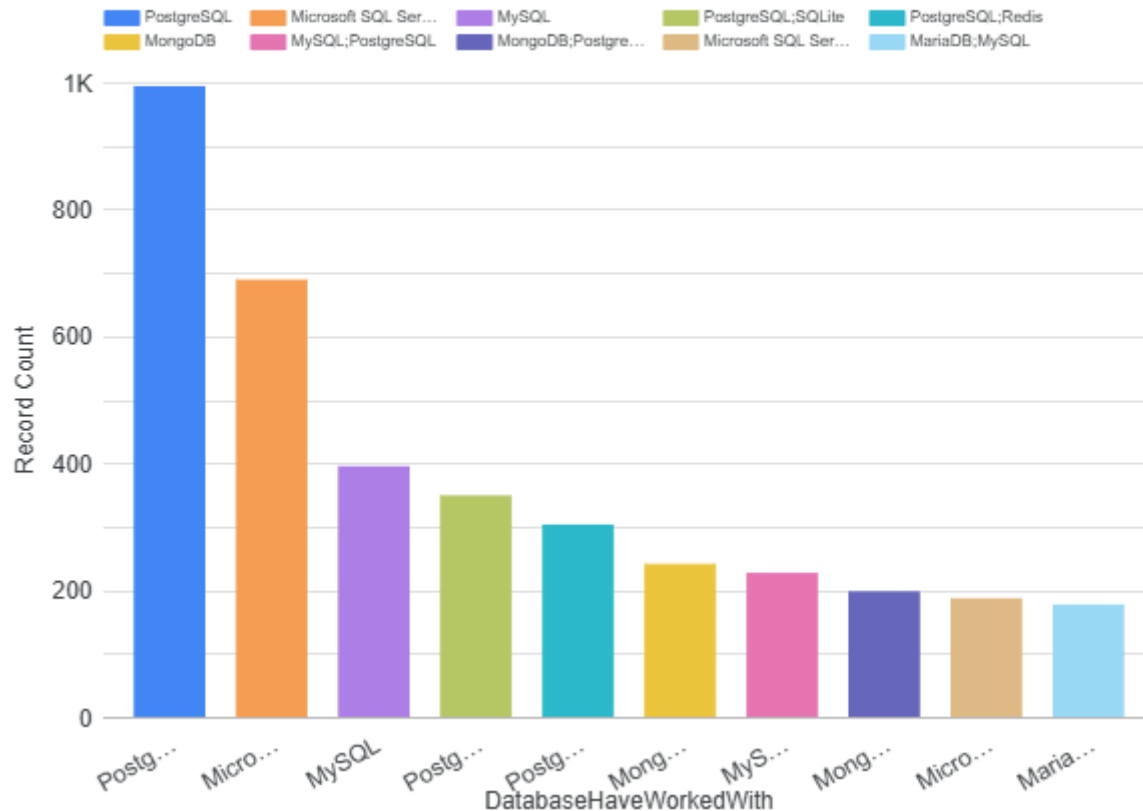
Implications

- Possible a lot of developers migration to C#

DATABASE TRENDS

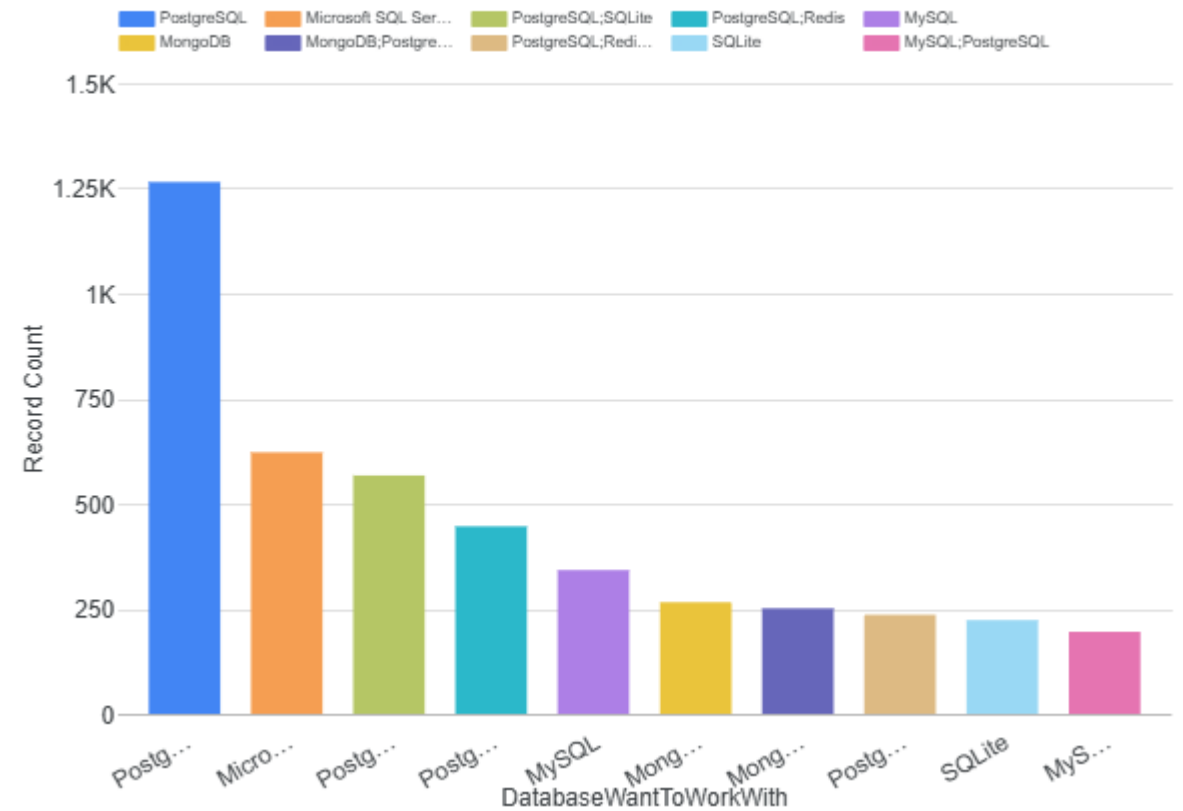
Current Year

Top 10 Databases Used



Next Year

Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL has significant growing and still keep as leading database
- Lack of interest in MySQL; PostgreSQL

Implications

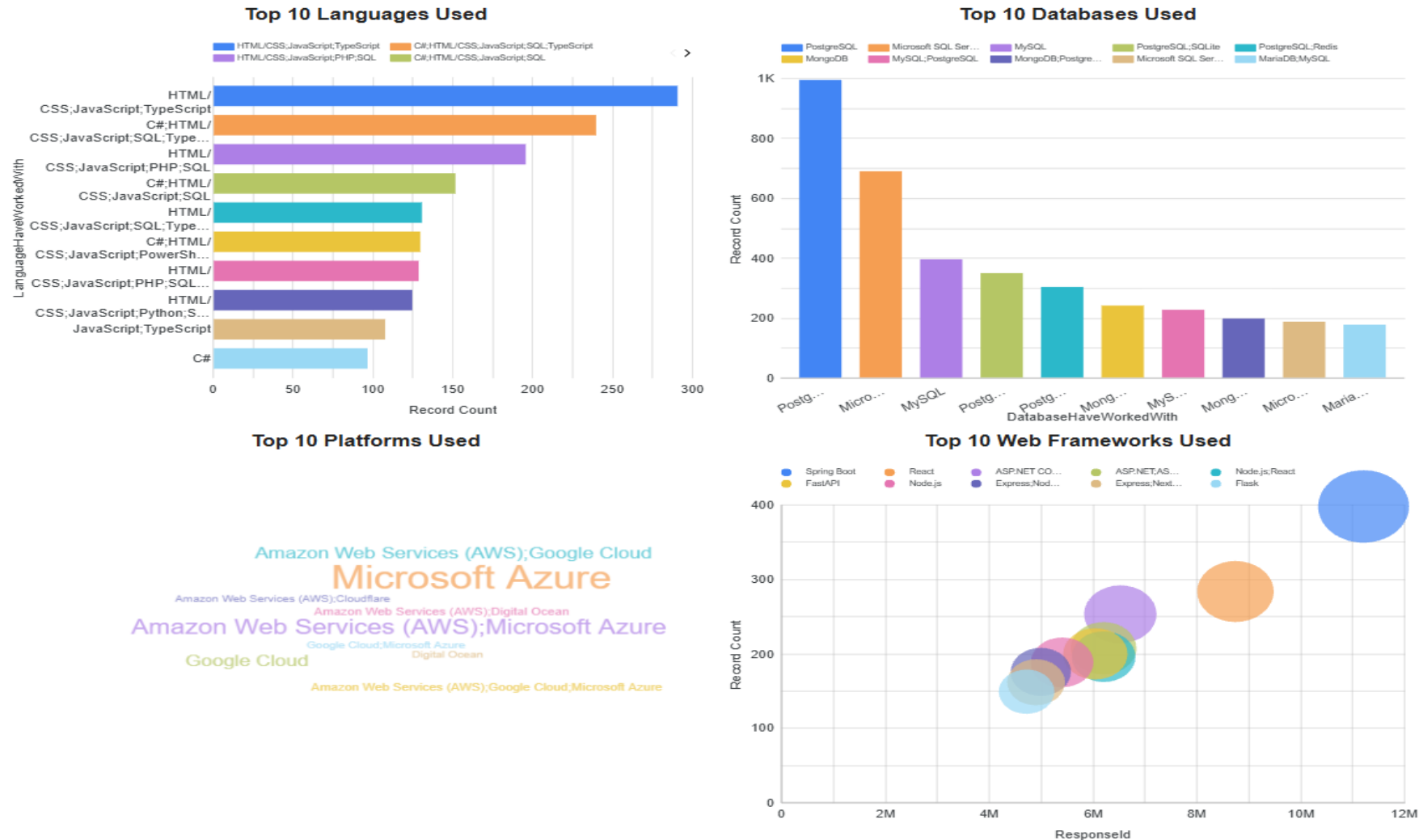
- Postgresql seems to be the most reliable database due to the increasing number of users



DASHBOARD

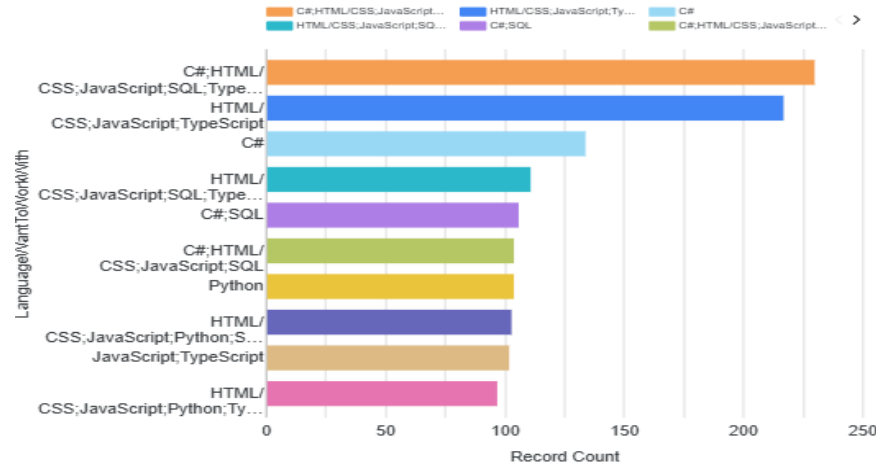


CURRENT TECHNOLOGY USAGE DASHBOARD

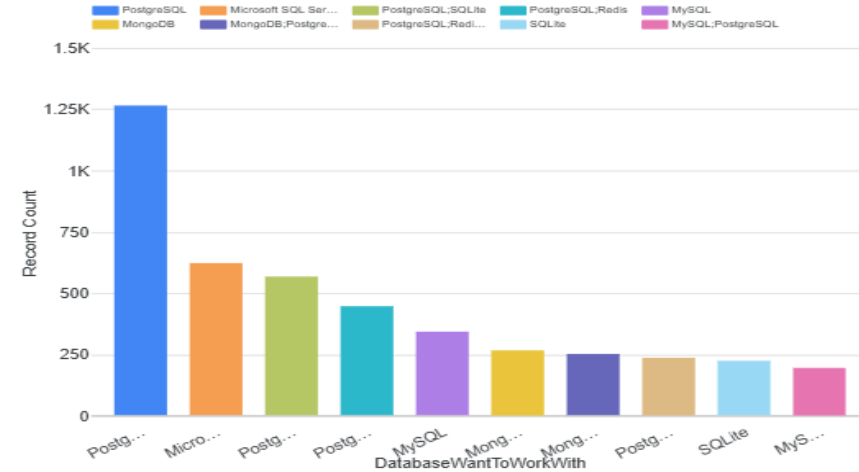


FUTURE TECHNOLOGY TRENDS DASHBOARD

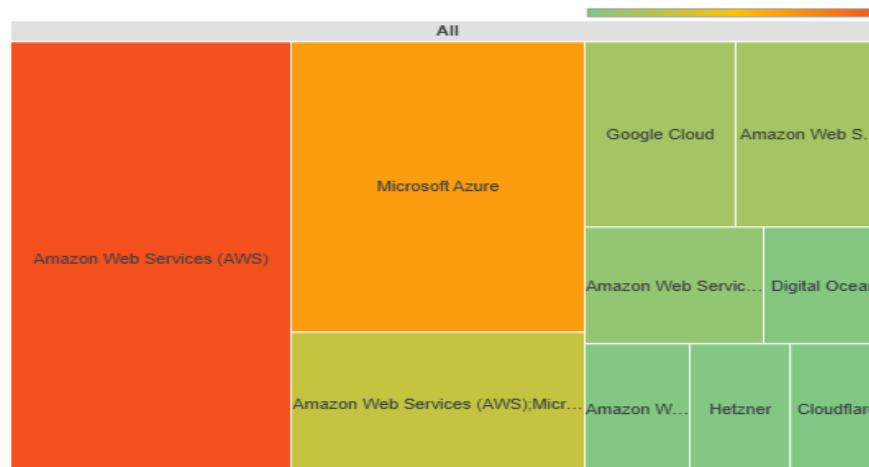
Top 10 Languages Desired Next Year



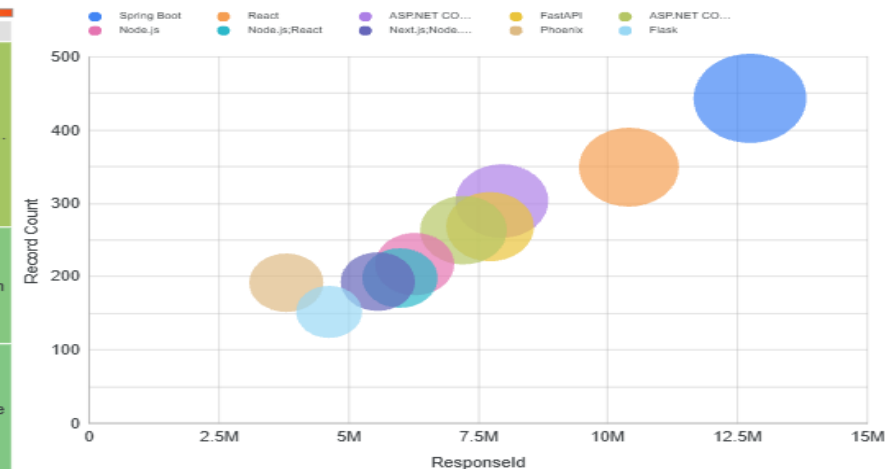
Top 10 Databases Desired Next Year



Top 10 Desired Platforms

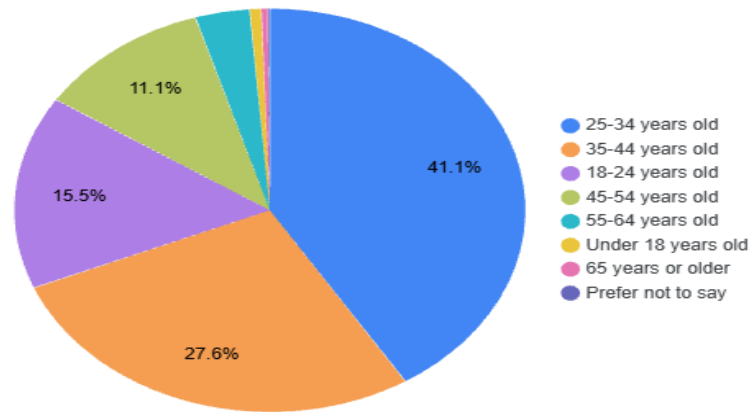


Top 10 Desired Web Frameworks

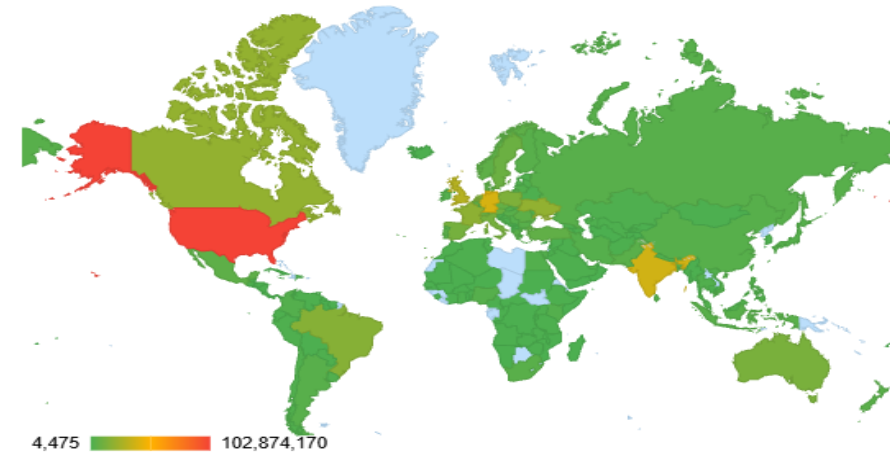


DEMOGRAPHICS DASHBOARD

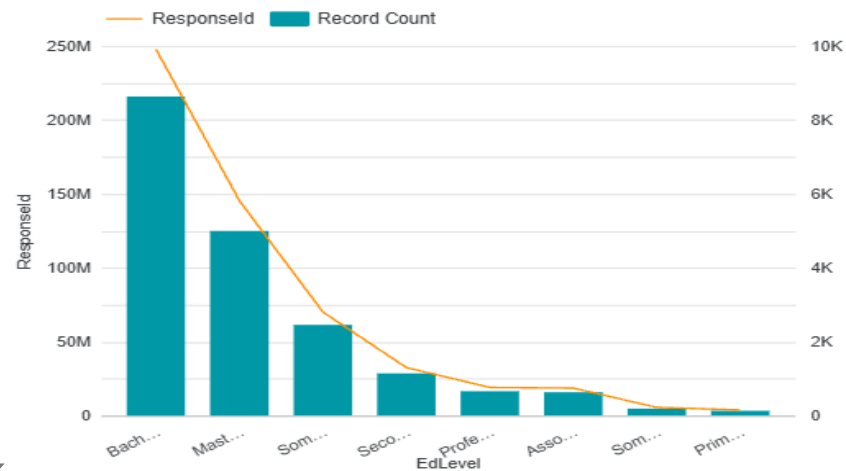
Respondents by Age



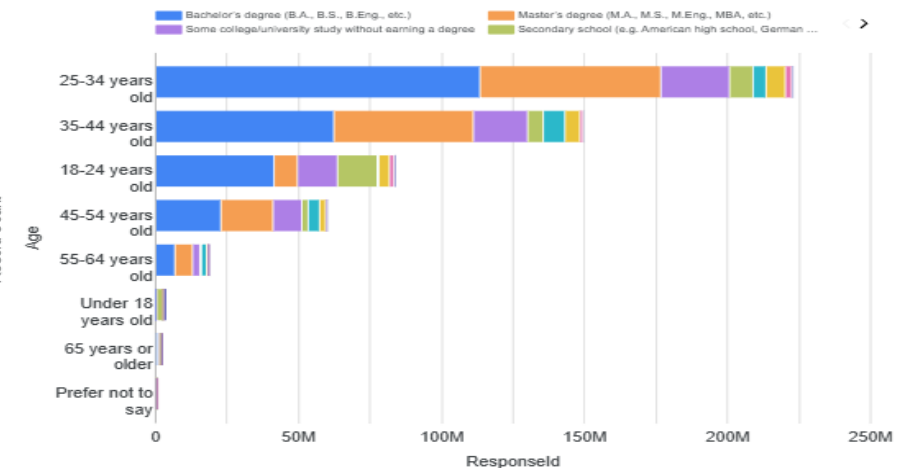
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- PostgreSQL has significant growing and still keep as leading database
- Most developers are around 25-34 years old
- Developer mostly located in developed countries

Implications

- Possible a lot of developers migration to C#
- Postgresql seems to be the most reliable database due to the increasing number of users

CONCLUSION



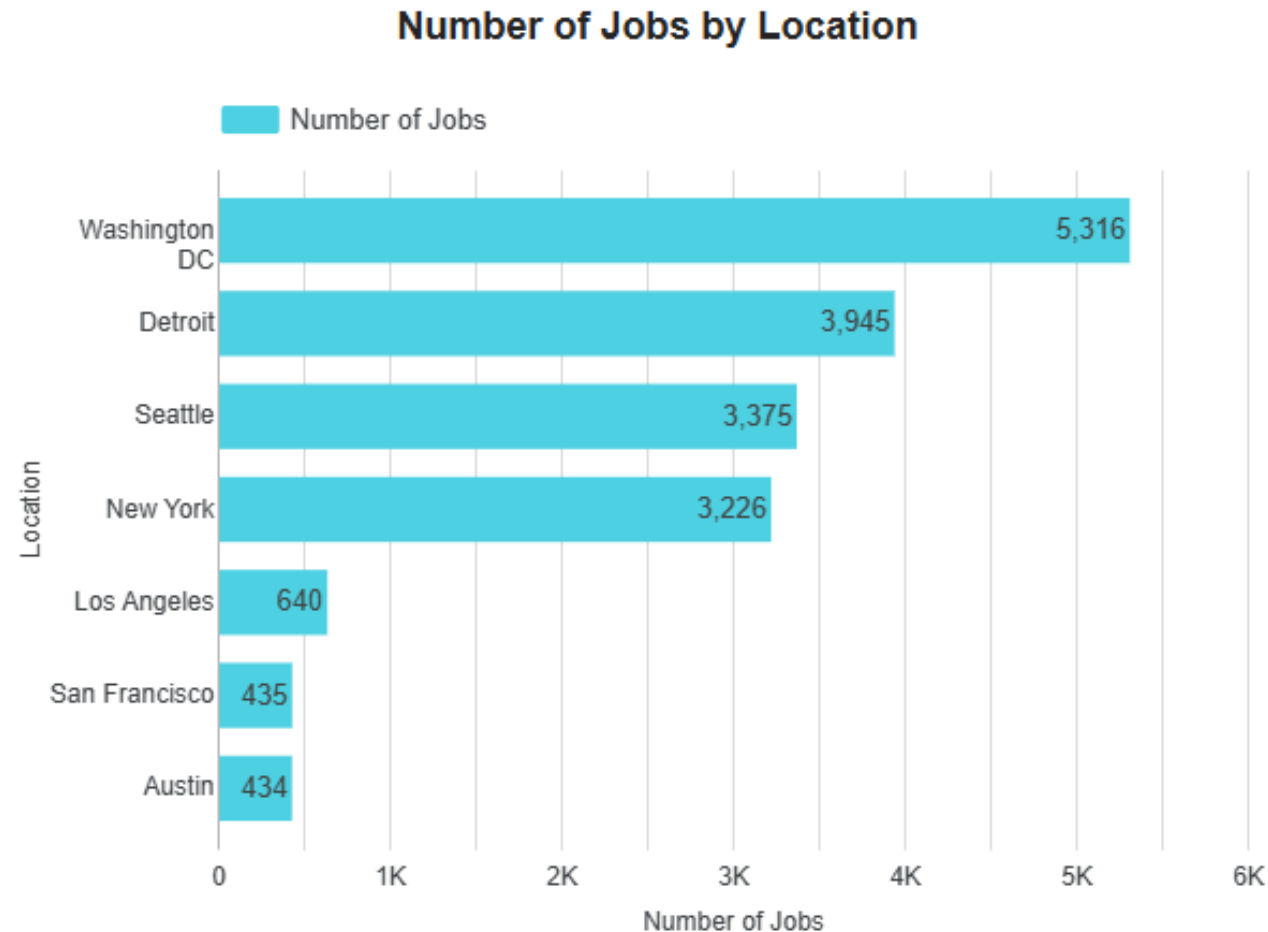
- Efforts are needed to expand access to this job market in developing countries.
- Developers tend to exhibit distinct and well-defined traits.
- It's possible to gain valuable insights into the popularity trends of various tools, platforms, and programming languages.



APPENDIX



JOB POSTINGS



POPULAR LANGUAGES

